

Preparation, Analysis, and Characterization of Some Transition Metal Complexes—A Holistic Approach

Kristy M. Blyth, Lindsay R. Mullings, David N. Phillips,* David Pritchard, and Wilhelm van Bronswijk

Department of Applied Chemistry, Curtin University of Technology, Perth, Western Australia 6845;

*D.Phillips@info.curtin.edu.au

Transition-metal complexes are an interesting and exciting group of compounds having a wide range of reactivity, physical, and chemical properties (1–3). Previous authors have reported the application of single substances to investigate certain aspects of transition-metal complexes. Clareen et al. (4) reported an undergraduate laboratory project where amine complexes of copper and silver were synthesized and subsequently analyzed by volumetric and gravimetric procedures. Tris(ethylenediamine)cobalt(III) chloride sulfate has been used as a compound to unify inorganic, analytical, and physical chemistry teaching, in particular examining its optical properties (5). We exploit the properties of transition-metal complexes to consolidate inorganic and analytical chemistry teaching. We conduct a unit in the second semester of the second year of our undergraduate Applied Chemistry degree course to study transition-metal complexes. Our approach is somewhat different: a wide range of complexes are studied among the student cohort. Following relatively simple preparative techniques, the complexes are characterized by a variety of instrumental methods, including infrared and UV–vis spectroscopy, magnetochemistry and conductance. In addition, the complexes are chemically ana-

lyzed using not only analytical techniques experienced in a previous analytical chemistry unit (6, 7), but also more advanced methods. Each student is required to analyze three elements in the allocated complex.

Experiment

Preliminary Work

Two weeks prior to the commencement of this project, the students are each allocated a separate transition-metal complex from List 1. The allocation of compounds is based on results from inorganic–analytical chemistry units studied in the previous semester. The so-called Group A complexes, which are inherently more challenging to analyze but relatively easy to prepare, are reserved for the high-achieving students, while the Group B complexes, which can be more laborious to prepare but are relatively easier to analyze, are allocated to the remainder of the class.

In the week prior to the commencement of the laboratory work, and while the previous activity of the semester is being completed (8), the students are given instruction on the use of Chemical Abstracts to gather information on transition-metal complexes. This includes use of the Formula Index (hard copies) and the SciFinder (computerized search). This allows the students to trace the method (or methods) that can be used to prepare the complex and gain information on previously published information on their complex.

Preparation of the Transition-Metal Complexes

Six laboratory sessions, each consisting of five hours of laboratory class time, is devoted to the practical component of this program. The first one or two weeks are generally taken up in the preparation of the transition-metal complex. The preparation of the Group B compounds are found in preparative inorganic chemistry textbooks (9–12), and the specific references are included in List 1. Note that the oxalato complexes are light sensitive, especially complex 12 and must be stored in a dark bottle or foil wrapped to avoid decomposition. The Group A compounds are examples of Schiff-base complexes, where the ligand is first produced by condensation of an aldehyde with an amine, followed by complexing the ligand with the metal ion. These complexes are relatively easy to prepare and readily precipitate from solution when an alcoholic solution of the ligand is added to an aqueous solution of the metal acetate or nitrate. The preparation of a typical complex from Group A is given in the Supplemental Material.^W

Students are required to calculate the scale required to prepare about 10 g of the complex, which is sufficient for all the physical and chemical testing. The yield of compound is compared with the published data.

List 1. Transition-Metal Complexes Studied by the Students

Group A (Schiff-base complexes)

1. Bis-[N-benzylsalicylideneiminato]nickel(II)
2. Bis-[N-benzylsalicylideneiminato]copper(II)
3. Bis-[N-2-hydroxyethylsalicylideneimine]nickel(II)
4. Bis-[N-2-hydroxyethylsalicylideneimine]copper(II)
5. Bis(salicylidene)ethylenediaminenickel(II)
6. Bis(salicylidene)ethylenediaminenickelcopper(II)

Group B

7. $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$ (ref 11, p 393)
8. $[\text{Co}(\text{NH}_3)_5\text{SO}_4]\text{Br}$ (ref 9, p 70)
9. $[\text{Co}(\text{NH}_3)_5\text{NO}_2]\text{Cl}_2$ (ref 11, p 394)
10. $[\text{Co}(\text{NH}_3)_5\text{ONO}]\text{Cl}_2$ (ref 9, p 67)
11. $\text{K}_3[\text{Cr}(\text{C}_2\text{O}_4)_3] \cdot 3\text{H}_2\text{O}$ (ref 12, Vol. 1, p 35)
12. $\text{K}_3[\text{Mn}(\text{C}_2\text{O}_4)_3] \cdot 3\text{H}_2\text{O}$ (Ref 10, p 1470)
13. $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3] \cdot 3\text{H}_2\text{O}$ (ref 12, Vol. 1, p 36)
14. $\text{K}_3[\text{Co}(\text{C}_2\text{O}_4)_3] \cdot 3\text{H}_2\text{O}$ (ref 12, Vol. 8, p 208)
15. $\text{K}_2[\text{Cu}(\text{C}_2\text{O}_4)_2] \cdot 2\text{H}_2\text{O}$ (ref 12, Vol. 6, p 1)
16. $[\text{Cr}(\text{en})_3]\text{Cl}_3 \cdot 3\text{H}_2\text{O}$ (ref 12, Vol. 13, p 233)
17. $[\text{Ni}(\text{en})_3]\text{Cl}_2 \cdot 2\text{H}_2\text{O}$ (ref 12, Vol. 6, p 200)
18. *trans*- $[\text{Co}(\text{en})_2\text{Cl}_2]\text{Cl}$ (ref 12, Vol. 2, p 222)

NOTE: C_2O_4 is oxalate and en is ethylenediamine.

Chemical Analysis

As previously mentioned, every student is required to chemically analyze three elements in their transition-metal complex. The calculation of the quantity of compound required for each determination is the responsibility of the student; however, the laboratory supervisor may provide guidance where considered necessary. The complexes in Group A are analyzed for their nitrogen content by the Kjeldahl method, the carbon content by a Leco carbon measurement, and the metal ion by flame atomic absorption spectrometry (FAAS). This group of complexes is somewhat more challenging in terms of chemical analysis since the complexes need to be digested in concentrated sulfuric acid prior to the determination. The determination of the metal ions in the complexes by FAAS analysis is an area that we have extensively studied using a program previously reported (8). The complexes are digested with concentrated nitric acid prior to FAAS analysis, and this method yields excellent results for the metal-ion content of the complexes. We have used ICP-OES analysis to cross-check the FAAS analytical data.

Complexes 7–10 of Group B are analyzed for their NH_3 content by the Kjeldahl method, the metal ion by FAAS, and the halide counter anion by conductance titration with standard silver nitrate. The trisoxalato complexes 11–15 are analyzed for the two metal ions by FAAS and the oxalate content determined by the Leco method. The ethylenediamine complexes 16–18 are analyzed for their nitrogen content by the Kjeldahl method, the metal ion by FAAS, and the chloride counter anion by conductance titration with standard silver nitrate. The empirical formula of the complex is deduced from the chemical analysis. In the cases of hydrated complexes, the student submits a sample of the complex to the laboratory technician for analysis of the water content by thermogravimetry, to ensure that the empirical formula of the compound may be correctly calculated. A brief description of the FAAS, Kjeldahl, and Leco procedures are given in the Supplemental Material.^W

Magnetochemistry

The magnetic moment of the transition-metal complex is determined by the Gouy technique. The calibration constant of the instrument is determined using mercury tetrathiocyanatocobaltate(II) as the standard reference material. Following the application of the diamagnetic correction (9), the magnetic moment may be calculated. The magnetic moment may then be correlated with the spin-only values calculated on the basis of octahedral, tetrahedral, or square planar stereochemistries and high-spin or low-spin d-electron configurations as applicable (13). Determination of the magnetic moment of the complex also allows students to identify the stereochemistry where there may be possible alternatives, in particular in the Schiff-base complexes. These may be square planar, tetrahedral, or even octahedral owing to solvent complexation, and the magnetochemistry of the nickel complexes 1, 3, and 5 are particularly good examples.

Infrared Spectroscopy

The infrared spectra of all the complexes are obtained using the KBr disc method. The spectra may be interpreted as a function of the bonding in the complex. In cases such as the nitro complex 9 and the nitrito complex 10, the stretching and bending vibrations are readily identifiable and different for the bonding of the ligand to the central metal. The preparation of complexes 1–6 may be followed by collecting spectra of the carbonyl compound and the amine, and observing the disappearance of the $\text{C}=\text{O}$ and $\text{N}-\text{H}$ peaks respectively, and the appearance of the $\text{C}=\text{N}$ peak in the Schiff base. Those requiring more intricate detail on the azomethine peak in such complexes should also record the spectra as a Nujol mull. The formation of the transition-metal complex is monitored by the disappearance of the OH peak in the Schiff base. Nakamoto's textbook is used as the reference source of spectra (14).

UV-Vis Spectroscopy

All students record the UV-vis spectrum of a 0.01 M solution of their complex between 320 and 900 nm. The importance of this measurement is to relate the peak(s) wavelength(s) to the d-orbital splitting and the color of their complex. The pentamminecobalt(III) complexes 7–10 clearly illustrate the spectrochemical series of increasing ligand strength $\text{Cl}^- < \text{SO}_4^{2-} < \text{ONO}^- < \text{NH}_3 < \text{NO}_2^-$, by the shift of the ${}^1\text{T}_1 \leftarrow {}^1\text{A}_1$ peak near 500 nm to shorter wavelengths. In the case of the Schiff-base complexes, the ligands mask most of the d–d transitions. It is thus not possible to identify the stereochemistry of the nickel(II) complexes by the method. Furthermore, the copper(II) complexes give only one peak and their stereochemistry also cannot be inferred from these spectra. We are not concerned with peaks in the UV region arising from transitions involving the ligands. Students with complexes in Group A use dichloromethane to dissolve the compound, while those with Group B complexes use aqueous solutions.

Conductance

The conductance of the transition-metal complex is related to the number of ions produced in aqueous solution. Typical conductance values for 0.001 M solutions of the following reference materials are provided for the students:

$[\text{Co}(\text{CO}_3)(\text{NH}_3)_4]\text{NO}_3$	2 ions	119 μS
$[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$	3 ions	280 μS
$[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$	4 ions	345 μS

Students measure the conductance of their complex following calibration of the conductance cell with 0.100 M KCl of molar conductivity 12.896 $\text{mS m}^2 \text{mol}^{-1}$ at 25 °C. The conductance value is not a highly accurate value, rather a value that confirms the number of ions present in aqueous solution. Students assigned to complexes of Group A are not required to measure the conductance since these compounds are only soluble in organic solvents and the solutions are non-conductive.

Reporting of all laboratory results on a weekly basis is handled as outlined in previous publications (6, 7) where students are required to hand in a duplicate sheet containing the details of a particular day's activities at the end of each laboratory session.

Hazards

Eye protection should be used at all times when carrying out the preparative and analytical procedures. Proper attention and caution should be applied when handling concentrated sulfuric, hydrochloric, and hydrobromic acids and 9 M sodium hydroxide. Special care should be exercised in the neutralization of the strongly acidic solutions by the 9 M NaOH in the Kjeldahl determinations.

Mercury tetrathiocyanatocobaltate(II) is toxic and dispensable hand gloves should be used in its handling. Dispensable hand gloves should also be used in handling the aldehydes and amines in the preparation of complexes of Group A.

The other reagents used in this exercise do not pose any significant hazard. MSDS sheets, which may be obtained from the Merck and Sigma Aldrich companies, must be consulted for each chemical prior to its handling. Some preparations suggest the use of ether in the final drying stage of the complex. However, apart from the $K_3[Mn(C_2O_4)_3] \cdot 3H_2O$ complex, acetone should be used as a substitute.

Equipment and Reagents

Apart from the Leco instrument and the Gouy balance, all equipment and reagents are to be commonly found in the laboratory.

Typical Student Data

The chemical analytical methods are all robust and student results are expected to fall within the following levels of accuracy for each particular method:

Conductometric titrations	±2%
FAAS analysis	±6%
Kjeldahl titration	±4%
Leco determination	±6%

Taking into account the realistic impact of the cumulative nature of these inaccuracies, typical results that are achieved for this experiment allow for full credit to be given to a student whose analytical data for say complex 1 from Group A fall within the extremes of $Ni_{0.9-1.1}C_{25.2-30.8}N_{1.8-2.2}$ for the analytes measured, and for complex 8 from Group B in the range $Co_{0.9-1.1}N_{4.5-5.5}Br_{0.9-1.1}$. We do not expect correspondingly high levels of accuracy in this program to those demanded in the purely analytical chemistry program (6, 7). The mark allocation in Table 1 reflects this approach where only some 25% of the marks are allocated to the chemical analysis component.

Table 1. Grading Scheme for the Laboratory

Section	Points
Introduction, includes evidence of reading literature	10
Experimental, includes details of instruments used, details of procedures used, and details of hazards	5
Chemical analysis, includes analytical data, interpretation of individual results, and calculation of empirical formula	15
Magnetic moment	6
IR and UV-visible spectra	6
Conductivity	3
Conclusion, includes the structure of the complex, together with a discussion including a table summarizing the data	5
(If references are used incorrectly, then 5 points or part thereof are deducted.)	
<i>Total</i>	<i>50</i>

Report

An introduction comprising about 500–1000 words must be submitted to the supervisor at the commencement of the third week of the program. This helps to spread the load on the student over the six-week period. The introduction must contain all the information that has been collected about the complex (or close analogues), including details of the chemistry of the central transition-metal ion, its tendency to form complexes, its stability, and the expected spectral and magnetic properties of the complex. It must be correctly referenced and, if of an unsatisfactory standard, will be returned for redrafting. Details of the preparative method for the complex and yield calculation are also submitted.

The full report is due in one week following the conclusion of the laboratory work. It must include the introduction, details of all experimental procedures including hazards, results from chemical and instrumental methods of analysis together with specimen calculations for each measurement, and a discussion of the way in which the ligand(s) is(are) bonded to the central atom in the complex. The climax of the report is the uniting of all results to show how they support the proposed structure of the complex, including a table that comprehensively summarizes these results alongside the expected values from the literature. The grading scheme is shown in Table 1. The total of 50 points translates to 60% of the laboratory score for the semester.

Student Comments

The students enjoy carrying out this exercise. Typical comments are “it integrates analytical, inorganic, and physical chemistry”, “a thorough investigation of transition-metal complexes”, “I had the opportunity of applying my previous analytical experience to this experiment”. The exercise is also highly recommended by the Department's Course Advisory Committee primarily comprised of chemists from industry.

Conclusion

The chemical and instrumental methods used in this study of transition-metal complexes provide a complete determination of their structure, bonding, and properties. It unites concepts of analytical, inorganic, and physical chemistry in a way that students may appreciate that these areas of chemistry are not worlds apart.

Supplemental Material

The preparation of a typical complex from Group A and a brief description of the FAAS, Kjeldahl, and Leco procedures are available in this issue of *JCE Online*.

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